

[Total No. of Questions: 09]
Uni. Roll No.

[Total No. of Pages: 02]

Program/Course: B.Tech.(Sem. 3rd)
Name of Subject: Mathematics-III
Subject Code: BSCS-101
Paper ID: 1104

Time Allowed: 03 Hours

EVENING

Max. Marks: 60

27 NOV 2019

NOTE:

- 1) Parts A and B are compulsory.
- 2) Part -C has Two questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately.
- 4) Scientific calculator is allowed.

Part -A

[Marks:02 each]

Q1.

- a) Write Cauchy-Riemann Equations in Cartesian form .
- b) Discuss Gauss Jordan method for solving three equations in x,y,z .
- c) Evaluate $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x^2$.
- d) Write algorithm for straight line fit of a curve.
- e) Check the correctness of the statement, "Mean of a binomial distribution is 3 and variance is 5."
- f) A machine produced 16 defective articles in a batch of 500. After overhauling it produced 3 defectives in a batch of 100. Has the machine improved?

Part -B

[Marks: 04 each]

Q2. Fit the curve $y = a + bx^2$ to the following data :

x	10	20	30	40	50
y	8	10	15	21	30

Q3. Expand $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ in the region $2 < |z| < 3$.

Q4. Solve the following equations by Gauss Elimination method :

$$2x + y + z = 10, 3x + 2y + 3z = 18, x + 4y + 9z = 16.$$

Q5. The nine items of a sample have the following values : 45, 47, 50, 52, 48, 47, 49, 53, 51.

Does the mean of these differ significantly from the assumed mean 47.5? ($t_{0.05} = 2.31$)

Q6. If on an average 1 vessel in every 10 is wrecked, find the probability that out of 5 vessels expected to arrive, at least 4 will arrive safely .

Q7. If $f(z) = u + iv$ is an analytic function and $u - v = e^x (\cos y - \sin y)$, find $f(z)$.

Q8. (a) i) Evaluate $\int_0^{2\pi} \frac{d\theta}{(17-8\cos\theta)}$ using contour integration.

ii) Find the bilinear transformation which maps the points $z = 1, i, -1$ on to the points $w = i, 0, -i$.

OR

(b) Using Cauchy Residue theorem, Evaluate $\int_c \frac{z^3}{(z-1)^4(z-2)(z-3)} dz$, where circle is $|z| = 5/2$.

Q9. (a) Fit a Poisson distribution to the following data and test the goodness of fit :

x	0	1	2	3	4
f	109	65	22	3	1

(Given that $\chi_{0.05}^2 = 7.815$)

OR

(b) i) Consider the random variable X which assigns the maximum of the number showing on a pair of dice. Find $E(X)$ and $\text{Var}(X)$.

ii) Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{x}{6} + k, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

Evaluate k and $P(1 \leq X \leq 2)$.
